

Scalability & Future Plan

Date	2 july 2026
Team ID	XXXXXX
Project Name	Flood Prediction
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Uses only historical weather data for prediction	Prediction accuracy may decrease if weather patterns change significantly	High
2	No real-time weather API integration	Users must manually enter weather parameters	High
3	Limited cloud deployment and notification support	No automatic SMS/Email alerts or large-scale deployment	Medium

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Supports a limited number of users on a local Flask server	Deploy on IBM Cloud or another cloud platform with load balancing to support many concurrent users.
2	Data Storage	Uses a CSV dataset for training and prediction	Migrate to a relational database such as MySQL or PostgreSQL to efficiently manage larger datasets and prediction logs.
3	Performance	Prediction is performed using a locally hosted ML model	Optimize the ML pipeline, use model caching, and enable GPU/cloud-based inference for faster predictions.
4	Security	Basic application security	Add user authentication, role-based access control, HTTPS, and encrypted data storage for secure access.

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
-------	-------------------------------	-----------------	-----------------

Phase 2	Deploy the application on IBM Cloud for remote access	1–2 Months	Improved accessibility, scalability, and availability.
Phase 3	Integrate real-time weather APIs and IoT sensor data	2-3 Months	More accurate and up-to-date flood predictions.
Phase 4	Add SMS, Email, and Mobile App notifications for early flood warnings	3-4 Months	Faster communication and better disaster preparedness.